LogiEdge

Phase 2 Report

Sensor pipeline, model validation and deployment evidence

Group 35 | AIML ZG535 - Machine Learning on Edge

BITS Pilani Work Integrated Learning Programmes
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1. Implementation scope

This report consolidates completed Components C-F, including the mandatory C2 experiment with statistics shifted by three standard deviations. It uses the same code, model identities and measured records as the final report. Results are from the completed laptop prototype; compiling this report did not rerun training, benchmarking, OTA or the live PSI experiment.

Runtime	Verified project environment
Host	Windows 11, i5-13420H, 8 cores / 12 logical processors, 15.7 GB RAM
Linux and containers	Ubuntu 24.04 on WSL2; Docker Desktop 4.97.0
Application base	Python 3.11-slim; Mosquitto 2.0.22; TensorFlow 2.14.1

1.1 Sensor pipeline and preprocessing (C1-C3)

Temperature is generated at 1 Hz from N(4.0, 0.3) degrees Celsius, and vibration RMS at 0.5 Hz from N(0.45, 0.05) g; the second argument denotes standard deviation. Door transitions carry timestamps. temp_drift adds 0.08 degrees per temperature reading; vibration faults use N(1.2, 0.15) g; combined activates both. Each 45-second smoke capture recorded 45 temperatures, 23 vibration summaries and three door events, with zero missed deadlines.

Both numeric streams use causal five-sample trailing averages. Complete 30-second windows advance by ten seconds; warm-up yields the first usable window at approximately 10-40 seconds. Each contains 30 temperature and 15 vibration values. Missing support, out-of-order data and wrong-truck readings are rejected rather than filled with invented values. Tests cover duplicates, causality, feature ordering and offline/incremental equivalence.

Exact six-input order: temperature mean; temperature population standard deviation; fitted temperature rate in degrees Celsius/min; vibration RMS; absolute vibration peak; Fisher excess kurtosis. Door state and scenario labels are not numerical model inputs.

2. Frozen statistics and model validation

2.1 Mandatory three-sigma experiment (C2)

A separate 600-second clean capture, seed 1001, provided 57 complete windows. Its six means and population standard deviations were saved as a 2 x 6 training_stats.npy array with feature-order and provenance metadata. No standard deviation required flooring. Runtime loads and transforms only; it never estimates replacement statistics from live readings [E1].

Feature	Frozen mean	Frozen std
temperature_mean	3.99115336	0.06553331
temperature_std	0.12580090	0.02366891
temperature_rate_c_per_min	-0.00616088	0.35798478
vibration_rms	0.44913708	0.01288813
vibration_peak	0.47931193	0.01664172
vibration_kurtosis	-0.78041345	0.43329297

The diagnostic replaces each mean with mean + 3 x original std while retaining the original standard deviation. Thus z_shifted = z_correct - 3 for each feature. M1 Keras predictions were evaluated on the same 59 held-out windows under both transformations; the saved production statistics remained unchanged [E2].

Normalisation condition	Accuracy	Change
Correct frozen statistics	100.00% (59/59)	Reference
Means shifted by +3 original std	100.00% (59/59)	0.00 percentage points

The measured change is zero. Strong separation of these synthetic scenarios does not establish that incorrect normalisation is safe for field data.

2.2 Dataset construction and validation (D1)

Scenario / label	Total	Train	Validation	Purged
20-min none / Normal	117	90	24	3
15-min temp_drift / Warning	87	66	18	3
15-min combined / Critical	87	67	17	3
Total	291	223	59	9

Chronological tails supply about 20% validation before purging. Nine windows are removed to separate overlapping raw-sample support. Training-only conditioning is folded into the first layer of the 6-32-16-3 ReLU MLP (803 parameters). Its weights are selected by training loss, without passing held-out validation to fit. All three TFLite variants classify 59/59 correctly; the confusion matrix is [[24,0,0],[0,18,0],[0,0,17]], with true rows and predicted columns [E2-E4].

3. Optimisation and five-metric benchmark

M1 is the 5,416-byte FP32 baseline. M2 uses full signed INT8 input/output and 223 unique training-only calibration rows, exceeding the required 200. M3 uses PolynomialDecay whole-unit gates, then physically removes 11/32 and 6/16 hidden units. Removing 17/48 units gives 35.42%, the whole-unit approximation to 35%, and a compact 6-21-10-3 model with 400 parameters. Both converted graphs pass integer-only audits [E3-E4].

Variant	Mean ms	p95 ms	KB	Accuracy %	Proxy mJ
M1 FP32	0.001496	0.001684	5.416	100.00	0.005632
M2 INT8	0.002161	0.002342	3.552	100.00	0.008154
M3 structured INT8	0.002109	0.002225	3.216	100.00	0.007986

Table 1. Decimal KB; invoke-only timings. Ten warm-ups are excluded, followed by 200 timed calls per model. Built-in CPU kernels, one thread, no default delegates; variant order was rotated. All 600 raw timing samples were independently checked [E5].

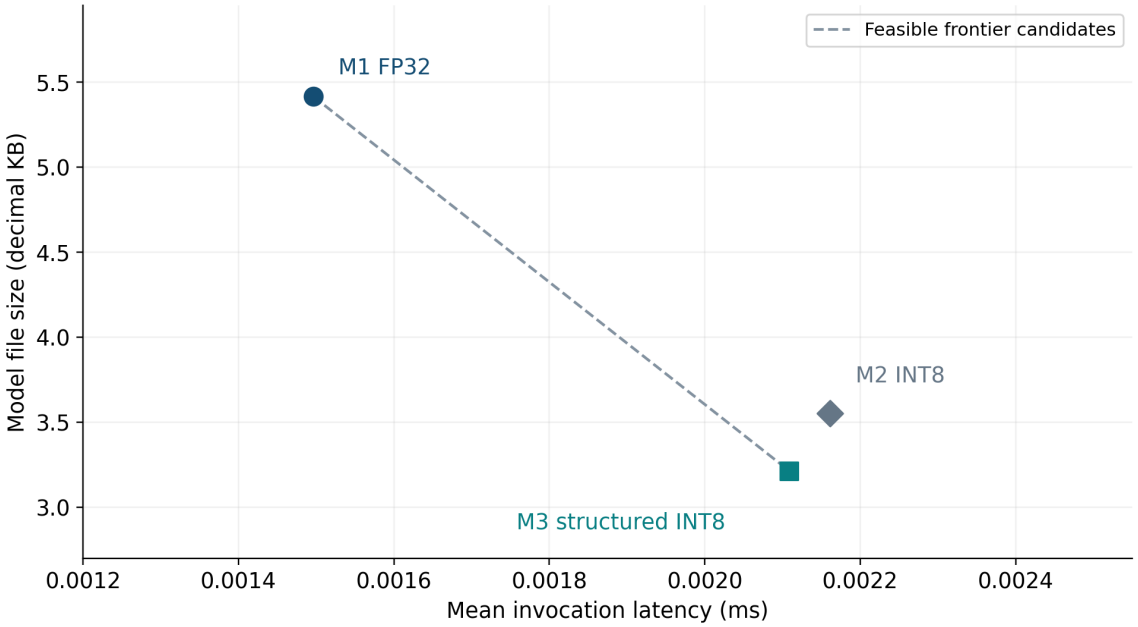


Figure 1. The feasible latency-size frontier contains M1 and M3. M3 dominates M2 on these measured results; a smaller INT8 file was not faster than FP32.

The energy estimate is a proxy: 45 W x process CPU% / (100 x 12) x mean invocation time in ms, giving mJ. CPU utilisation comes from a separate five-second probe. It is not measured battery energy or whole-node power.

Recommendation. M1 is selected for the verified laptop CPU runtime: lowest mean, p95 and energy proxy, with Critical recall 17/17 = 100%. M3 is the smallest candidate (40.62% smaller than M1). Approximately 40 seconds of initial collection leaves about 50 seconds of the illustrative 90-second budget; microsecond model invocation is only one part of that budget. Pi 5 RAM, storage and sustained ARM64 latency need target profiling.

Both INT8 models clipped 35 validation feature values in 35 rows, while remaining correct. The small synthetic holdout, including only 17 Critical windows, supports the assignment point thresholds rather than a field safety guarantee.

4. Runtime, offline recovery and cached OTA

4.1 Docker inference and durable delivery (D2)

The inference image uses Python 3.11-slim and installs dependencies before the final model COPY. MODEL_PATH selects the artifact without rebuilding: the existing M1 image successfully loaded registered M3 INT8 in the saved test. The service publishes to logibridge/trucks/{truck_id}/inference and records model/statistics identities with the result [E6].

Independent raw-temperature rules use the assignment setpoint of 4 degrees Celsius: deviation greater than one degree produces Warning and greater than three produces Critical. Operational severity also accounts for the latest model class. This is necessary because the prescribed temp_drift training scenario remains labelled Warning even after its unbounded temperature crosses a physical Critical limit.

During the controlled combined run, saved status reported a local physical Critical event at 35.08 seconds and first model Critical at 40.16 seconds. When the operations service was unavailable, seven uplink events stayed pending, survived inference service recreation and later all received ACKs [E7]. MQTT handling and local logging continued independently of uplink availability. This backend-interruption test is a local proxy for an outage, not a measured rural cellular trial.

4.2 Actual model-layer cache demonstration (D2)

Registry-payload quantity	Saved result
Changed filesystem layers	1: final model COPY; earlier build steps cached
Cold target image per truck	347,760,955 bytes
Cached update per truck	13,994 bytes = 1,642 layer + 12,352 config/manifest
85-truck cold / cached projection	29,559.681175 MB / 1.189490 MB
Payload reduction / projected cost saved	99.995976% / Rs 2,955.849169

These figures count unique compressed registry objects, with the prior image cached in the update case. Protocol, security, retries and billing overhead are excluded. They are fleet projections from measured local objects [E6].

4.3 Separate 280 KB strategy comparison (E3)

Strategy for 85 trucks	Model-only payload	Cost at Rs 0.10/MB
Full replacement	23.8 MB	Rs 2.38
Canary 10, then promote 75	2.8 + 21.0 MB	Rs 0.28 + Rs 2.10
Shadow; previous model cached	23.8 MB new weights	Rs 2.38 + unspecified overhead

Recommend canary to limit simultaneous exposure and permit rollback before promotion. It does not reduce the completed model-only fleet cost. Shadow adds duplicate compute and diagnostic traffic; full replacement exposes all trucks at once. Production rollout and security remain proposed.

5. Live PSI monitoring and recovery (E1)

The primary score is **P(Normal)**, not maximum class confidence. The frozen reference uses 300 clean model windows. Four bins cover [0,0.25), [0.25,0.50), [0.50,0.75) and [0.75,1.0], including one in the last bin. An empty-bin floor of 1e-6 and renormalisation keep the statistic finite.

```
PSI = sum((current_i - reference_i) x ln(current_i / reference_i))
```

The last 100 distinct inference events are evaluated every 60 seconds. The buffer and frozen reference remain intact across the combined injection and clean restoration. The completed experiment used actual MQTT model probabilities; replay independently reproduced all 24 full-window PSI evaluations [E8].

Live measurement	Saved result
Duration / actual MQTT model outputs	2,460.018 monotonic seconds / 242
Clean full-window baseline PSI	0.000000
First PSI after injection above threshold	0.425487 (> 0.25)
Injection to alert	60.000262 seconds (within 300 seconds)
Recovery PSI / clean restore to recovery	0.000000 (< 0.10) / 1,079.001771 seconds
Reference unchanged / missed sensor deadlines	True / zero

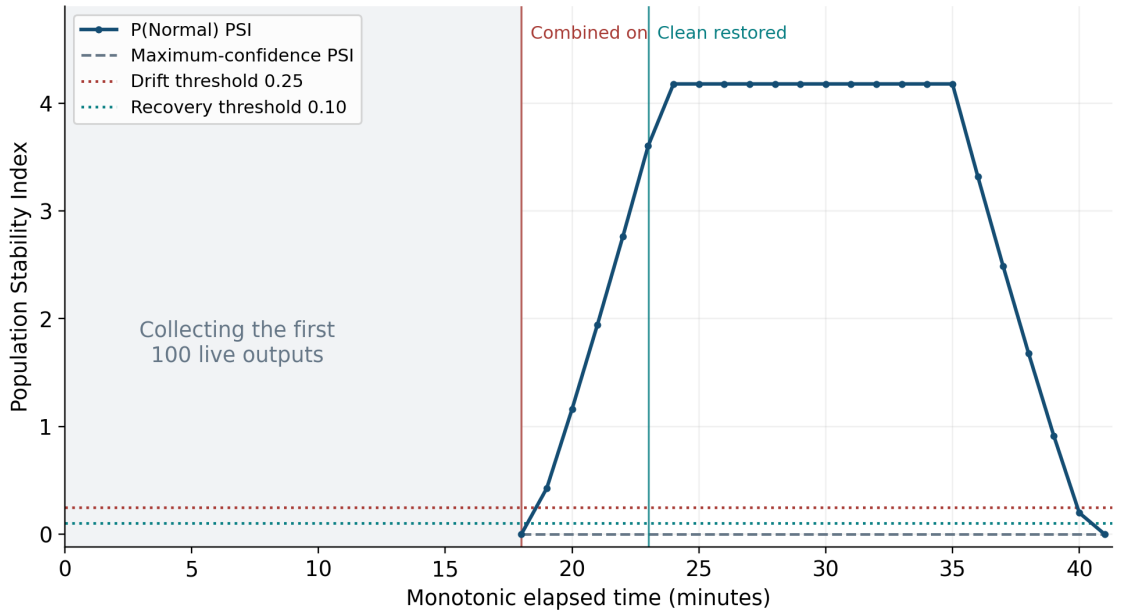


Figure 2. Saved live measurements from run 2291a861fa424c138da62d062037e160. The remembered abnormal outputs explain slow recovery of the rolling buffer.

Maximum-class-confidence PSI stayed zero because both Normal and Critical predictions remained confident. The observed shift is an operating-regime and output-distribution change; it does not prove concept drift or loss of classification accuracy. PSI monitoring is separate from the immediate cargo alarm path.

Clock guard v2 uses monotonic elapsed time and separately checks suspend / processing gaps. The passing run logged 28 UTC adjustments (about -67.96 seconds cumulative), a maximum loop gap of 0.270 seconds and no interruption requiring review. Historical negative UTC-derived alert durations are excluded from latency evidence.

6. Seven-task deployment and integration

6.1 Ansible deployment (E2)

The root playbook logibridge_deploy.yml contains exactly seven tasks and disables automatic fact collection. State-aware helper actions avoid stopping or restarting an unchanged deployment simply to report a zero change count [E9].

Task	Operation
1	Create /opt/logibridge with the required ownership and permissions
2	Copy the selected model as model.tflite
3	Copy the matching frozen reference_dist.json
4	Stop running inference if a deployment update is needed
5	Pull the pinned image from the local registry
6	Start inference with the specified environment and volumes
7	Wait 15 seconds and verify running service and artifact identities

Run	OK	Changed	Failed	Unreachable
First recorded run	7	5	0	0
Unchanged second run	7	0	0	0

Container ID and StartedAt are identical across the two saved runs, independently supporting the no-restart claim. Verification checks the loaded model, normalisation and reference. The runtime override uses read-only deployment assets and preserves the durable edge-state volume.

6.2 Fresh post-deployment check

A subsequent 45-second clean run, seed 8001, acknowledged 45 temperatures, 23 vibration summaries and three door events, with no missed deadlines. It produced a fresh M1 Normal prediction and operational Normal state with no clipped features. The saved status has zero pending uplinks; acknowledged uploads and remote unique events both total 72 [E10].

6.3 Technical difficulty and remaining work

An early live experiment stopped because the clock check treated UTC adjustments as an interruption. Separating monotonic elapsed intervals from calendar timestamps allowed an uninterrupted, auditable rerun. Failed and intentionally stopped runs remain distinguishable from the passing run; they are not presented as completed evidence.

Before field adoption, add real sensor validation, operationally consistent labels, wider independent Critical cases, representative INT8 calibration coverage, target memory/power measurements and ARM64 profiling. Add authenticated encrypted uplink, signed immutable model/reference versions, rollback checks and explicit clock-domain metadata. The proposed canary fleet rollout has not been executed.

7. Evidence index and reproducibility

Paths below are relative to the shared project repository root. They refer to the original saved measurements, not newly generated execution claims. The scope and required thresholds come from the LogiEdge mini-project brief; pipeline terminology follows M4_Sessions7_8_SlideDeck.pptx.

[E1] Frozen normalisation

`data_pipeline/training_stats.npy`
`data_pipeline/training_stats.meta.json`

[E2] Dataset, M1 and three-sigma result

`training/dataset/dataset_manifest.json`
`training/models/baseline_20261102T165253Z_9f6488/metrics.json`

[E3] M2 full INT8

`training/models/ptq_20261102T170433Z_cfa235/metrics.json`
`training/models/ptq_20261102T170433Z_cfa235/integer_audit.json`

[E4] M3 structured compaction and INT8

`training/models/structured_20261102T171917Z_3c7e14/metrics.json`
`training/models/structured_20261102T171917Z_3c7e14/structured_pruning_audit.json`

[E5] Raw timing and five-metric benchmark

`optimisation/results/benchmark_20261102T174448Z_44017b/benchmark_results.json`
`optimisation/results/benchmark_pointer.json`

[E6] Model-layer update and MODEL_PATH

`reports/ota_20261103T112924Z_eOcY7H/ota_results.json`

[E7] Offline persistence and delivery

`reports/offline_after_restart.json`
`reports/after_sync.json`
`reports/normal_recovery.log`

[E8] Completed live PSI and independent audit

`reports/psi_live_20261104T044934Z_L9uSpi/audited_summary.json`
`reports/psi_live_20261104T044934Z_L9uSpi/live_results.json`

[E9] Actual two-run deployment

`logibridge_deploy.yml`
`reports/ansible_20261104T062754Z_C85o1J/ansible_results.json`

[E10] Fresh post-deployment status

`reports/post_deployment_status.json`

Protected artifact identities

Normalisation SHA256

`e96f950a1d222bd0087084c119b8545a29f96c600699ab70601c2663569ff251`

Dataset SHA256

`a905545a17f88a240dddc4313a26f15f1668b59b791bc63749e805acbafd7034`

Passing PSI run / frozen reference SHA256

`2291a861fa424c138da62d062037e160`
`217a8a4b21f4969b506927dcfb6958a131cb76f62ac0548e97aa34037c64dad3`

These identities connect the reports to the saved artifacts. Model, statistics, reference distributions and original measurement files were not changed to prepare these phase reports.